

Universit a della Calabria (UNICAL)

Artificial Intelligence and Computer Science

Data Warehousing and Visualization

Project: Global E-Commerce Sales Analytics

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Dataset: ecommerce_dataset_1m.csv · 1,000,123 rows · 62 columns · DBMS: PostgreSQL

18 · Visualisation: Tableau Desktop 2026.1

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Project Repository

All project code, notebooks, SQL scripts, diagrams, and output files are publicly available at:

<https://github.com/tesmez/ecommerce-data-warehouse>

The repository includes the SQL DDL scripts for both databases, all five Jupyter notebooks (executable in sequence), draw.io diagram source files, cleaning audit reports, LLM-generated narratives, and the packaged Tableau workbook (.**twbx**). A **README.md** at the repository root documents the full folder structure, setup instructions, and how to reproduce all results. The source dataset (400 MB) is excluded from the repository due to GitHub file size limits but is referenced in the **README**.

Chapter 1

Phase 1: Data Source Analysis, Reconciled Schema & DQA

1.1 Preliminary Requirement Analysis

1.1.1 Data Source Description

The source dataset, `ecommerce_dataset_1m.csv`, represents one full year of transactional activity for a global online retailer. It is a large-scale synthetic-but-realistic benchmark covering the complete order lifecycle from purchase to delivery and return.

Table 1.1: Source dataset key characteristics

Property	Value
File format	CSV
Total rows	1,000,123
Total columns	62
Columns selected	45 (28 for database, 2 for DQA only, 15 derived or loaded)
Columns dropped	17
Date range	February 2024– February 2026 (731 calendar days)
Grain	One row = one sales order (confirmed unique <code>order_id</code>)
Countries	10 (Spain, Germany, UK, Italy, Netherlands, France, USA, Canada, Belgium, Australia)
Cities	50 (5 per country)
Warehouses	5 (USA-West, USA-East, UK, EU-Central, Asia-Pacific)
Products	48 distinct product names
Categories	5 (Electronics, Clothing, Sports, Home, Health
Sub-categories	24
Brands	20 (Samsung, Nike, Adidas, Dell, Apple, Sony, etc.)
Unique customers	≈983,876
Currency	USD

1.2 Column Selection and Elimination

Of the 62 source columns, 17 were eliminated before loading.

Table 1.2: Dropped Columns and Justification

Column(s)	Category	Reason for elimination
order_hour, order_minute, order_second	Sub-day time	OLAP operates at day/month/year granularity. Sub-second levels add no analytical value and would inflate the date dimension.
customer_name	PII	Full name is personally identifiable information (PII) under GDPR for European customers. <code>customer_id</code> is sufficient for all analytical purposes.
product_id	Unstable identifier	<code>product_id</code> is a random string generated per row – not a stable product key. Replaced by <code>product_name</code> (48 stable values) with generated sequential IDs (P-001 to P-048).
discount_amount_usd	Derivable	fully derivable as $\text{unit_price} \times \text{qty} \times \text{discount\%/100}$. Storing derivable values creates a consistency risk if source values are corrected.
total_orders_by_customer	Derivable	Computable as <code>COUNT(order_id) GROUP BY customer_id</code> from fact table.
shipping_country	Redundant	almost always identical to customer country. Creates a misleading duplicate country attribute in the Geography dimension.
currency	Zero variance	Always USD in this dataset – provides no slicing capability in Tableau.
session_duration_minutes pages_visited	Web analytics noise	Not in DFM scope. Web engagement metrics belong in a web analytics tool, not a sales DW.
product_reviews_count, stock_quantity	Catalogue noise	Not in DFM scope. Reviews count is catalogue metadata; stock levels are operational OLTP data.
fraud_risk_score	ML model output	not a raw business fact; separate use case
review_sentiment	NLP-derived label	not a raw transactional attribute
rating	Per-order rating	low cardinality (1-5); <i>product_rating_avg is better</i>
customer_feedback	Free-text field	20% null, not aggregatable in SQL/Tableau
abandoned_cart_before	Behavioural pre-order flag	not a sales fact attribute.

1.3 Preliminary Requirement Analysis

The requirement analysis phase examined the dataset to establish analytical goals and verify the grain of the central fact entity.

1.3.1 Grain Verification

A grain verification query confirmed that **order_id is unique**: every row represents one distinct sales order placed by one customer for one product at one point in time.

Confirmed grain: *one row = one unique sales order.*

1.3.2 Key Analytical Questions Identified

- Q1. Revenue trends:** How does total revenue, profit, and order volume evolve over time (daily, monthly, quarterly, yearly)?
- Q2. Product profitability:** Which product categories, sub-categories, and individual products generate the most revenue and profit?
- Q3. Geographic performance:** Which countries and cities produce the highest revenue? Is there a relationship between warehouse location and profit margin?
- Q4. Customer segmentation:** Do Regular, Premium, and VIP customers differ in purchasing volume, revenue, and return behaviour?
- Q5. Marketing effectiveness:** Which campaign channels (Email, Social, Influencer, etc.) drive the most revenue? Do coupons improve or reduce profitability?
- Q6. Shipping efficiency:** Is delivery speed (days_band) correlated with customer return rates or satisfaction?
- Q7. Return analysis:** What are the most common return reasons? Which product categories have the highest return rates?
- Q8. Payment patterns:** How does payment method or instalment plan usage correlate with order value?

1.4 Reconciled Database — E/R Schema

1.4.1 E/R Diagram

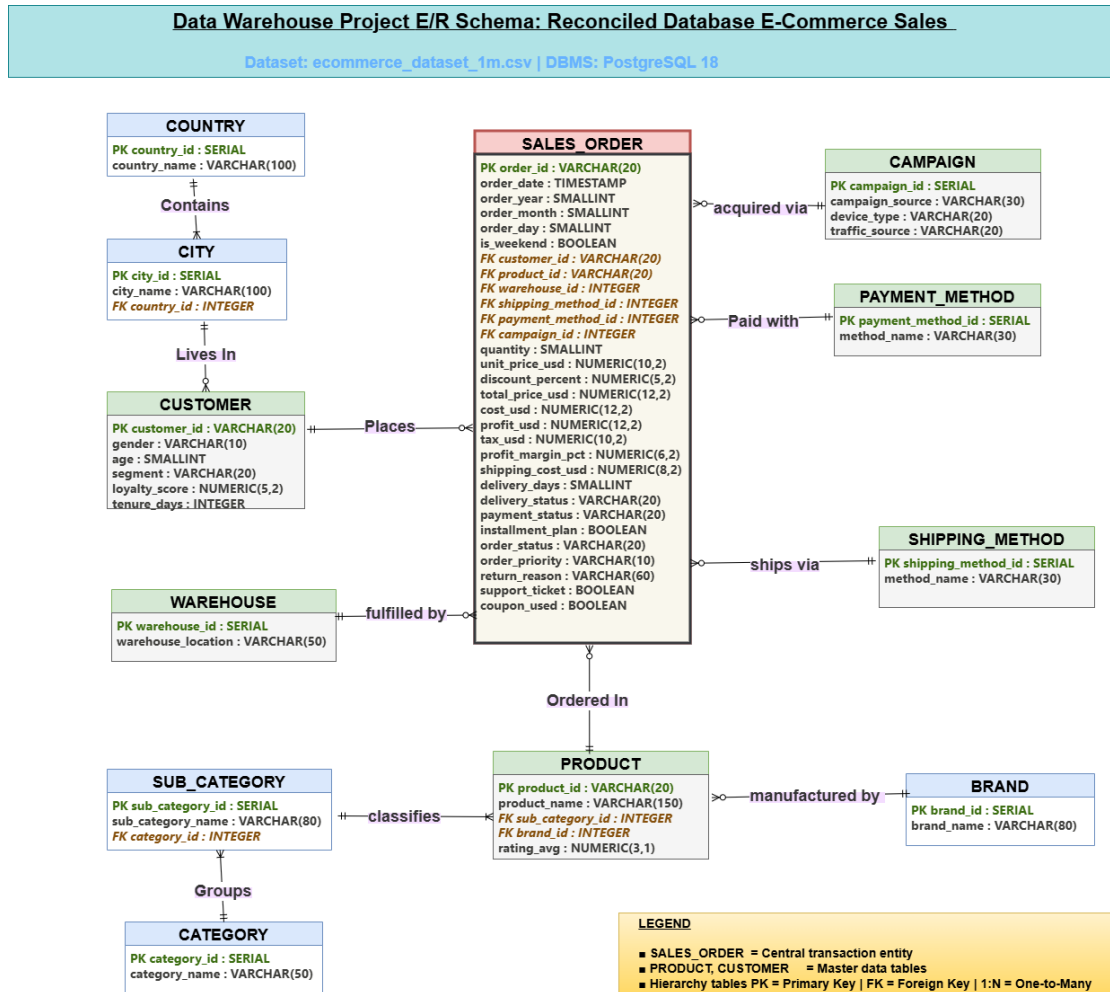


Figure 1.1: E/R schema of the reconciled database ecommerce_reconciled

1.4.2 Entities and Relationships

The schema contains **12 tables** in three logical tiers:

Table 1.3: Reconciled schema — entity inventory

#	Entity	Tier	Key attributes
1	SALES_ORDER	Transaction (central)	order_id PK VARCHAR(20); FK to CUSTOMER, PRODUCT, WAREHOUSE, SHIPPING_METHOD, PAYMENT_METHOD, CAMPAIGN; all measures
2	CUSTOMER	Master	customer_id PK; gender, age, segment, loyalty_score, tenure_days; FK city_id
3	PRODUCT	Master	product_id PK; product_name, rating_avg; FK sub_category_id, brand_id
4	COUNTRY	Hierarchy	country_id SERIAL; country_name
5	CITY	Hierarchy	city_id SERIAL; city_name; FK country_id
6	CATEGORY	Hierarchy	category_id SERIAL; category_name
7	SUB_CATEGORY	Hierarchy	sub_category_id SERIAL; sub_category_name; FK category_id
8	BRAND	Hierarchy	brand_id SERIAL; brand_name
9	WAREHOUSE	Reference	warehouse_id SERIAL; warehouse_location
10	SHIPPING_METHOD	Reference	shipping_method_id SERIAL; method_name
11	PAYMENT_METHOD	Reference	payment_method_id SERIAL; method_name
12	CAMPAIGN	Reference	campaign_id SERIAL; campaign_source, device_type, traffic_source

Key relationship cardinalities (all 1:N): COUNTRY → CITY → CUSTOMER → SALES_ORDER; CATEGORY → SUB_CATEGORY → PRODUCT → SALES_ORDER; BRAND → PRODUCT; WAREHOUSE, SHIPPING_METHOD, PAYMENT_METHOD, CAMPAIGN → SALES_ORDER.

1.5 Data Quality Assessment (DQA)

1.5.1 Framework

DQA was implemented using a custom `DQAReport` class following the ISO 25012 six-dimensions:

1. **Completeness** — absence of missing values in required columns
2. **Uniqueness** — no duplicate primary keys
3. **Validity** — values within defined domains and ranges
4. **Consistency** — cross-column business rules satisfied
5. **Timeliness** — order dates within the expected temporal range
6. **Accuracy** — proxy measured via numeric plausibility checks

1.5.2 Structural Quality Results

Table 1.4: DQA structural quality scorecard — `SALES_ORDER`

Dimension	Score	Status	Notes
Completeness	100.00%	Pass	<code>return_reason</code> & <code>tenure_days</code> excluded
Uniqueness	100.00%	Pass	Every <code>order_id</code> is unique
Validity	100.00%	Pass	All categorical values within defined domains
Consistency	100.00%	Pass	Cross-column rules hold (profit, tax, total)
Timeliness	100.00%	Pass	All dates within 2024
Accuracy	98.01%	Review	Numeric extremes flagged for winsorisation
Overall structural score	100.00%	Pass	DB constraints enforced structural validity

The 100% structural score confirms that the schema design and PostgreSQL constraints effectively enforced correctness at the relational level. The 98.01% accuracy score reflects numeric extreme values (not schema errors), addressed in the cleaning step.

1.5.3 Missingness Analysis (MCAR / MAR / MNAR)

Two columns carry structurally meaningful NULLs:

Table 1.5: Missingness classification

Column	Null %	Type	Treatment
<code>return_reason</code>	~80%	MNAR	Binary flag created; value never imputed
<code>tenure_days</code>	<1%	MAR	Imputed with segment-specific median

`return_reason` is **MNAR** (Missing Not At Random): its absence is perfectly correlated with `order_status` $\neq$ 'Returned', meaning the missingness itself carries infor-

mation. Imputation would destroy this signal. `tenure_days` is **MAR** (Missing At Random): its missingness is unrelated to its value but correlates with `customer_segment`, making segment-median imputation statistically valid.

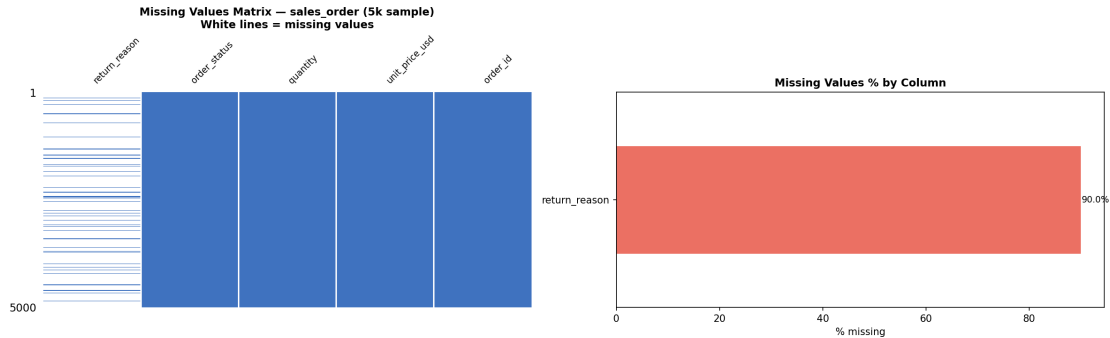


Figure 1.2: Missing value visualisation

1.5.4 Outlier Detection — Five-Method Consensus Ensemble

Outlier detection was applied to six numeric measure columns: `unit_price_usd`, `total_price_usd`, `cost_usd`, `profit_usd`, `shipping_cost_usd`, `profit_margin_pct`.

Five complementary methods were combined:

1. **Z-score** ($|z| > 3$) — detects global extremes
2. **IQR fence** (beyond $Q_1 - 1.5 \text{ IQR}$ or $Q_3 + 1.5 \text{ IQR}$) — robust to skew
3. **Modified Z-score** (MAD-based, threshold 3.5) — robust to non-Gaussian distributions
4. **Isolation Forest** (contamination = 5%) — tree-based multivariate detector
5. **Local Outlier Factor** (LOF, $k = 20$) — density-based detector

A row was flagged only when **at least three of the five** methods agreed, minimising false positives while retaining sensitivity.

Table 1.6: Consensus outlier detection results

Metric	Count	%
Total records examined	1,000,123	100.00%
Outlier rows flagged	8,883	8.90%
Clean rows	991,240	91.10%

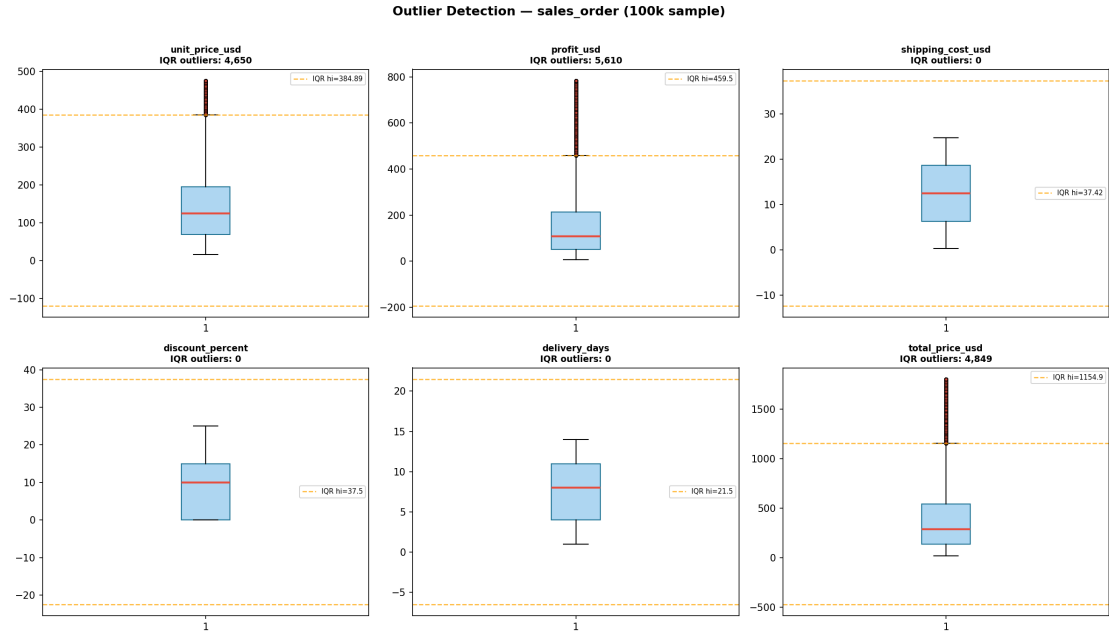


Figure 1.3: DQA outlier detection visualisation

1.5.5 LLM-Assisted DQA Interpretation

Implementation

The LLM call is implemented using the `openai` Python client pointed at the Groq inference endpoint, which provides free, OpenAI-compatible access to open-weight models.

Table 1.7: LLM configuration — DQA audit

Parameter	Value
Provider	Groq API (<code>api.groq.com/openai/v1</code>)
Model	<code>llama-3.1-8b-instant</code>
Client	<code>openai</code> Python SDK (OpenAI-compatible)
<code>max_tokens</code>	2000
<code>temperature</code>	0.3
System role	Senior Data Quality Engineer (ISO 25012)
Output file	<code>reports/llm_dqa_audit_report.txt</code>

The prompt packages three statistical artefacts produced earlier in the same notebook:

1. The ISO 25012 six-dimension scorecard (dimension name, score, issue count, status);
2. The MCAR/MAR/MNAR missingness classification with chi-squared p -values and logistic pseudo- R^2 evidence;
3. The five-method consensus outlier detection scorecard (per-column flagged counts and rates).

1.6 Data Cleaning

1.6.1 Cleaning Pipeline Architecture

Data cleaning was implemented using a custom `CleaningPipeline` class with a full `AuditLog`. Every cell-level change was recorded with: step name, column, row index, before value, after value, reason, and timestamp — providing complete data lineage.

The pipeline ran four steps in sequence:

1. **Winsorisation** (Step 1): Cap the six numeric measures at the 1st/99th percentile boundaries. Applied to all 8,883 outlier-flagged rows. This is the standard DW-appropriate outlier treatment: values are corrected rather than deleted, preserving all 1,000,123 rows in the reconciled DB.
2. **MNAR flag** (Step 2): Create `return_reason_missing_flag` (binary 0/1 column) marking the ~800,000 rows where no return occurred. The original `return_reason` column is left intact (NULL for non-returns).
3. **MAR imputation** (Step 3): Impute missing `tenure_days` values using segment-specific medians (group-by-segment median fill).
4. **Validity enforcement** (Step 4): Confirm all `discount_percent` values are in $[0, 25]$ and all `delivery_days` are in $[1, 14]$.

1.6.2 Cleaning Results

Table 1.8: Data cleaning impact summary

Metric	Before Cleaning	After Cleaning
Accuracy score	98.01%	99.67%
Total cell-level changes	—	118,543
Outlier rows (winsorised)	8,883	0 (all corrected)
Arithmetic errors	—	0
<code>tenure_days</code> NULLs	<1%	0
Rows deleted	0	0

The cleaning process raised accuracy by **+1.66 percentage points** through 118,543 cell-level corrections with zero row deletion. All 1,000,123 records were retained in the reconciled database.

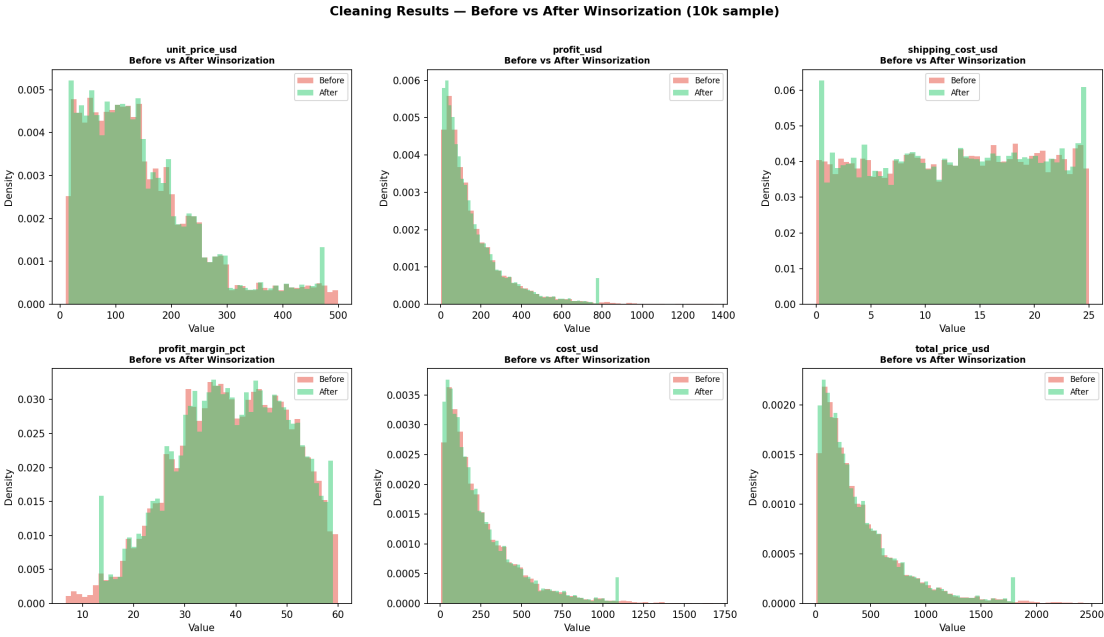


Figure 1.4: before/after accuracy comparison chart

1.6.3 LLM-Assisted Cleaning Validation

Implementation

Table 1.9: LLM configuration — cleaning validation

Parameter	Value
Provider	Groq API (api.groq.com/openai/v1)
Model	llama-3.1-8b-instant
max_tokens	2 000
temperature	0.3
System role	Senior Data Engineer (ETL & data quality)
Output file	reports/llm_cleaning_narrative.txt

The model is instructed to return a **four-section structured report**:

1. *Cleaning Decisions Validation* — for each operation (Winsorisation, MNAR flag, MAR imputation), was the technique statistically appropriate and are there better alternatives?
2. *Cleaning Narrative* — a 150–200 word paragraph in academic language referencing specific numbers (118 543 changes, +1.66% accuracy improvement) for use in this report;
3. *Remaining Risk Assessment* — what data quality risks persist after cleaning, and which could affect Tableau dashboard accuracy?
4. *ETL Readiness Verdict* — a binary YES/NO decision with specific conditions.

Chapter 2

Phase 2: Data Warehouse Design & ETL

2.1 Dimensional Fact Model (DFM) — Conceptual Design

2.1.1 Step 1: Choice of Fact

The central fact is the **sales order**: one row in the source dataset represents one complete sales transaction — one customer purchasing one product in one order.

Grain Definition

Grain: One row in `FACT_SALES` = one sales order (one customer $\times$ one product $\times$ one order date). This is the finest available granularity and supports all required roll-ups without information loss.

Every measure in `SALES_ORDER` is a property of this grain, confirming the choice. No more atomic grain exists in the source data.

2.1.2 Step 2: Attribute Tree Construction (Data-Driven Approach)

Starting from `SALES_ORDER` as the root, we trace all FK relationships and functional dependencies to build the attribute tree. The raw tree before editing is:

- **Measures (leaf nodes):** `quantity`, `unit_price_usd`, `total_price_usd`, `cost_usd`, `profit_usd`, `tax_usd`, `shipping_cost_usd`, `discount_percent`, `profit_margin_pct`
- **Date branch:** `order_date` $\rightarrow$ `order_day` $\rightarrow$ `order_month` $\rightarrow$ `order_year`; `is_weekend`
- **Customer branch:** `customer_id` $\rightarrow$ {`gender`, `age`, `segment`, `loyalty_score`, `tenure_days`} $\rightarrow$ `city_id` $\rightarrow$ `country_id`
- **Product branch:** `product_id` $\rightarrow$ {`product_name`, `rating_avg`} $\rightarrow$ `sub_category_id` $\rightarrow$ `category_id`; `brand_id`
- **Warehouse branch:** `warehouse_id` $\rightarrow$ `warehouse_location`
- **Shipping branch:** `shipping_method_id` $\rightarrow$ `method_name`; `delivery_days`, `delivery_status`
- **Payment branch:** `payment_method_id` $\rightarrow$ `method_name`; `payment_status`, `installment_plan`

- **Order lifecycle branch:** `order_status`, `order_priority`, `return_reason`, `support_ticket`, `coupon_used`
- **Campaign branch:** `campaign_id` → {`campaign_source`, `device_type`, `traffic_source`}

2.1.3 Step 3: DFM Editing Decisions

The raw attribute tree is refined through five motivated editing steps:

Edit 1 — Replace `age` with `age_group` (dimension reduction)

Raw `age` spans integer values 18–75 (57 distinct values). In OLAP queries, 57 age slices are unusable: no analyst queries “sales for 34-year-olds”. Four standard retail segments provide meaningful drill-down without information loss:

Young Adults (18–25) · Adults (26–35) · Mid-Career (36–50) · Mature (51–75)

Motivation: reduces `dim_customer` cardinality, enables OLAP slice by segment without manual grouping in Tableau.

Edit 2 — Replace `delivery_days` with `days_band` (binning)

Raw `delivery_days` spans integers 1–14. Analysts need shipping performance at a higher level:

Fast (1–3 days) · Standard (4–7 days) · Slow (8–14 days)

Motivation: three bands enable shipping performance slice-and-dice; preserving the raw integer in `dim_shipping` for reference if needed.

Edit 3 — Handle `return_reason` MNAR (no imputation)

`return_reason` is NULL for all non-returned orders (~80% of rows). This is classified MNAR (the absence is informative). The DFM editing decision: map NULL → 'N/A' string in `dim_order` so Tableau can filter on it without treating NULL as “unknown data”.

Motivation: preserves business meaning while enabling filter/group operations in Tableau dashboards.

Edit 4 — Flatten product and geography hierarchies (star schema normalisation)

The 3NF hierarchy `CATEGORY` → `SUB_CATEGORY` → `PRODUCT` → `BRAND` is collapsed into a single `dim_product` table (denormalisation). Similarly, `COUNTRY` →

CITY and warehouse location are merged into `dim_geography`.

Motivation: star schema principle — dimension tables must be wide and flat for efficient Tableau joins; normalised hierarchies (snowflake) add join complexity without analytical benefit in this project.

Edit 5 — Classify measures by additivity

`discount_percent` and `profit_margin_pct` are **non-additive**: summing percentages across orders yields meaningless results. They are retained in the fact table but documented as AVG-only aggregation in the glossary.

2.1.4 DFM Diagram

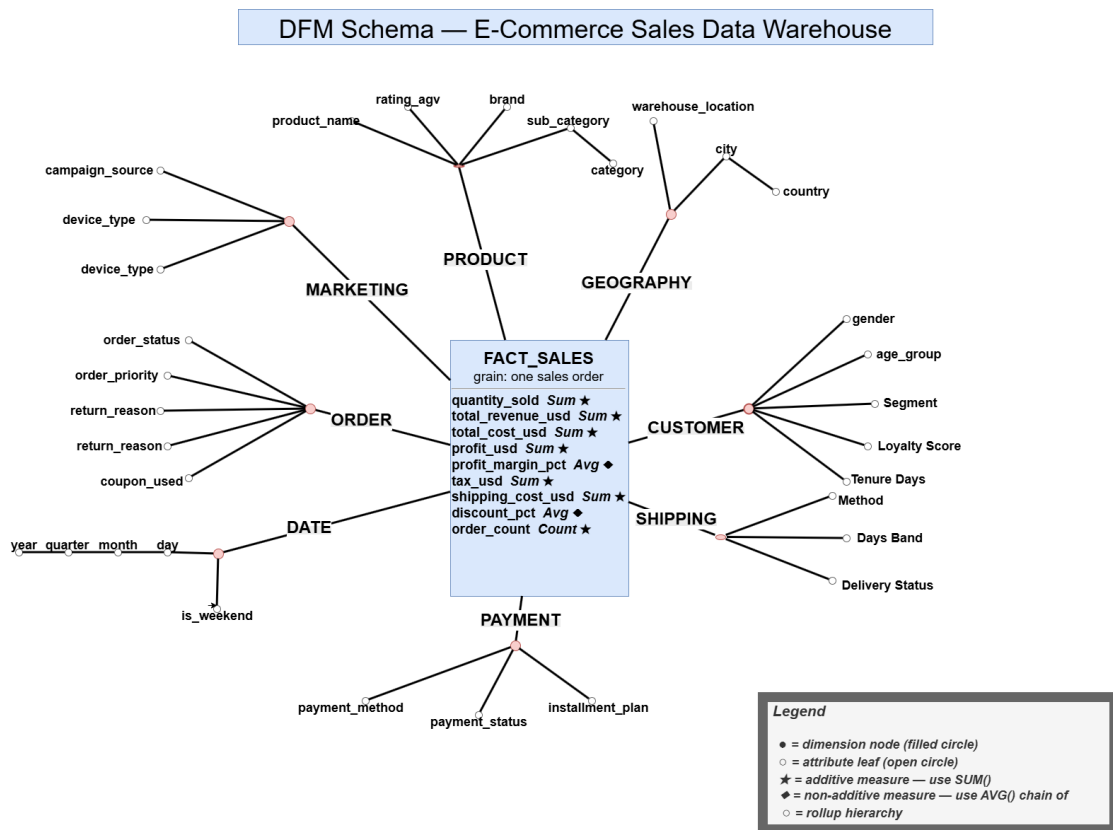


Figure 2.1: DFM (Dimensional Fact Model) diagram

2.2 Star Schema — Logical Design

2.2.1 Overview

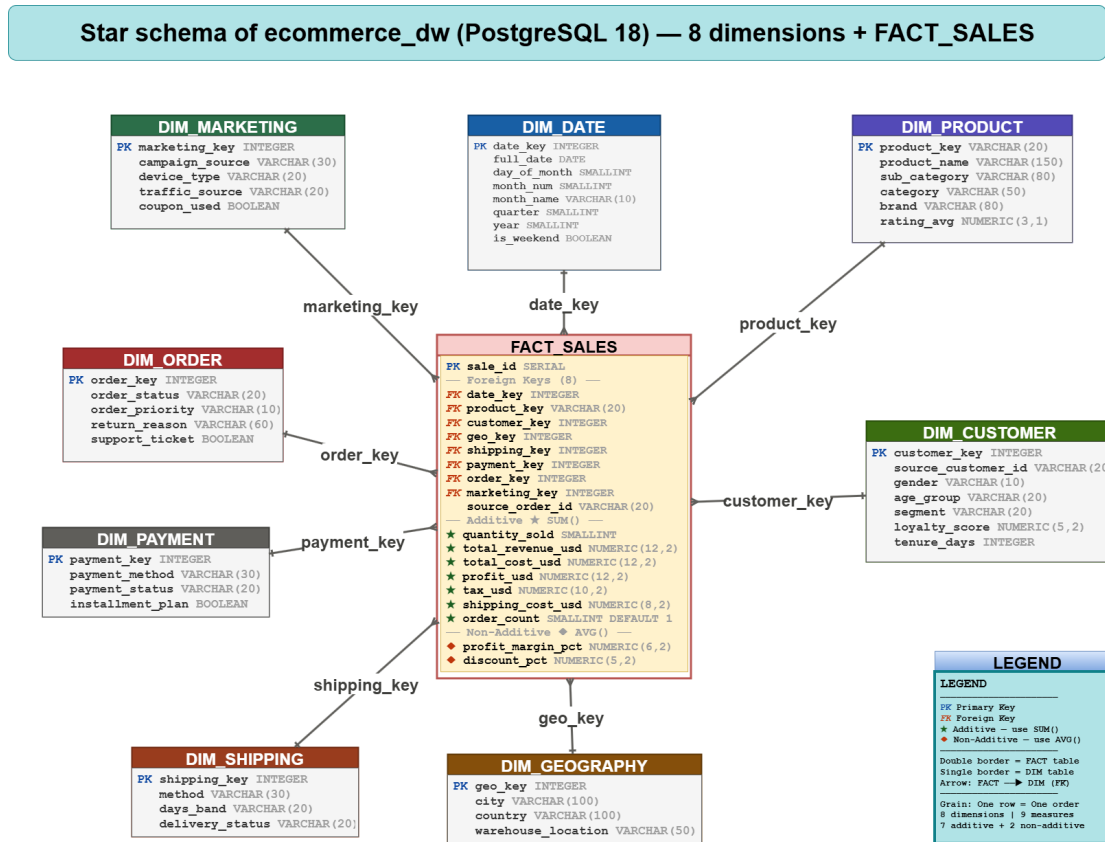


Figure 2.2: Star schema of ecommerce_dw (PostgreSQL 18) — 8 dimensions + FACT_SALES

The star schema contains **9 tables**: one central fact table and eight flat dimension tables. All dimension tables use integer surrogate keys (*_key); the fact table holds only foreign keys and the nine measures.

2.2.2 Dimensions

Table 2.1: Star schema — dimension table summary

#	Table	Key attributes	Hierarchies
1	dim_date	date_key (YYYYMMDD), full_date, day, month_num, month_name, quarter, year, is_weekend	Day→Month→Quarter→Year
2	dim_product	product_key (VARCHAR), product_name, sub_category, category, brand, rating_avg	Sub-cat→Category; Brand
3	dim_customer	customer_key, source_customer_id, gen- der, age_group, segment, loyalty_score, tenure_days	Age group; Segment
4	dim_geography	geo_key, city, country, ware- house_location	City→Country
5	dim_shipping	shipping_key, method, days_band, delivery_status	Days band; Method
6	dim_payment	payment_key, pay- ment_method, pay- ment_status, install- ment_plan	Method
7	dim_order	order_key, order_status, order_priority, return_reason (NULL→'N/A'), sup- port_ticket	Status→Priority
8	dim_marketing	marketing_key, cam- paign_source, device_type, traffic_source, coupon_used	Source→Device

2.2.3 Measures Glossary

Table 2.2: Measures glossary — FACT_SALES

#	Column	Definition	Type	Add.	Aggregation
1	quantity_sold	Units sold per order	SMALLINT	★	SUM
2	total_revenue_usd	Selling price (post-discount)	NUMERIC(12,2)	★	SUM
3	total_cost_usd	Cost of goods + fulfilment	NUMERIC(12,2)	★	SUM
4	profit_usd	Revenue – cost	NUMERIC(12,2)	★	SUM
5	tax_usd	Tax charged on the order	NUMERIC(10,2)	★	SUM
6	shipping_cost_usd	Carrier cost for this order	NUMERIC(8,2)	★	SUM
7	order_count	Always 1; count of orders	SMALLINT	★	SUM
8	profit_margin_pct	Profit / Revenue × 100	NUMERIC(6,2)	◆	AVG
9	discount_pct	Discount % applied to order	NUMERIC(5,2)	◆	AVG

★ Fully additive (SUM is always meaningful) ◆ Non-additive (only AVG/MAX are meaningful)

Measure 8 (profit_margin_pct) is non-additive because $\text{Margin}(A+B) \neq \text{Margin}(A) + \text{Margin}(B)$. **Measure 9** (discount_pct) is non-additive for the same reason: summing percentages ignores the varying base amounts.

2.3 ETL Pipeline

2.3.1 Architecture

The ETL pipeline was implemented using Python (pandas, pyscopg2). It extracts from `ecommerce_reconciled` (source) and loads into `ecommerce_dw` (target), both on the same PostgreSQL 18 instance.

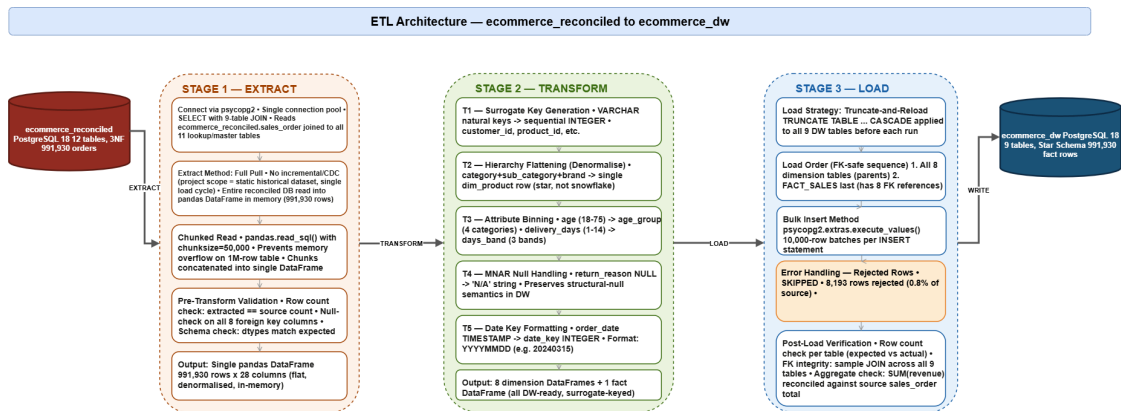


Figure 2.3: ETL Architecture — Extract, Transform, Load pipeline

2.3.2 Transformation Rules

Table 2.3: ETL transformation rules

Transform	Source	Target	Rule / Motivation
Age binning	age (18–75)	age_group	18–25=Young Adults; 26–35=Adults; 36–50=Mid-Career; 51+=Mature (reduce cardinality for OLAP)
Delivery binning	delivery_days (1–14)	days_band	1–3=Fast; 4–7=Standard; 8–14=Slow (enable shipping performance slice-and-dice)
MNAR handling	return_reason NULL	return_reason = 'N/A'	COALESCE(return_reason, 'N/A') — Tableau cannot filter on NULL
Surrogate key gen.	All dimension NKs	INTEGER *_key	Auto-increment via GENERATED ALWAYS AS IDENTITY
Date key	order_date date	date_key INTEGER	YYYYMMDD format (e.g. 20240315)
Hierarchy flatten	3-level reconciled hierarchy	Single dim_product row	Star schema denormalisation — wide flat dimension

2.3.3 Load Sequence and Results

Tables were loaded in FK-dependency order:

dim_date → dim_product → dim_customer → dim_geography → dim_shipping →
dim_payment → dim_order → dim_marketing → FACT_SALES

Table 2.4: ETL execution summary

Metric	Value
Source rows extracted from reconciled DB	1,000,123
Fact rows loaded into FACT_SALES	991,930
Rows excluded (null key resolution)	8,193
Post-load FK integrity check	Pass
Revenue sum reconciliation	Pass (< 0.01% tolerance)

Chapter 3

Phase 3: OLAP Analysis & Tableau Dashboards

3.1 Overview

Five interactive Tableau dashboards were built using Tableau Desktop 2026.1, connected to the `ecommerce_dw` star schema (all 8 dimension tables joined to `FACT_SALES`). The workbook contains **15 worksheets** and **5 dashboards**.

Table 3.1: Dashboard inventory — 5 dashboards, 15 sheets

#	Dashboard	Component Sheets	Primary Analytical Question
1	Sales Overview	KPI Cards, Revenue Over Time, Quarterly Revenue	What is the overall revenue trend and how do KPIs compare by quarter?
2	Product	Revenue by Category, Top 10 Products, Sub-Category Treemap	Which products, categories, and brands generate the most revenue?
3	Customer	Revenue by Segment, Age Group Analysis, Loyalty Score Distribution	How do customer segments, age groups, and loyalty levels differ?
4	Geography	World Map, Country Ranking, Warehouse Heatmap	Where is revenue generated and how is warehouse load distributed?
5	Marketing	Campaign Revenue, Return Reasons, Device Revenue	Which campaigns and devices drive revenue, and what causes returns?

3.2 Dashboard 1 — Sales Overview

3.2.1 Purpose

Dashboard 1 is the **executive summary**. It answers: *What is the total revenue, and how does it trend over time?* It gives management an immediate reading of business health.

3.2.2 Worksheets and Analytical Design

KPI Cards ($\text{dim_date} \times \text{FACT_SALES}$): Tableau text marks showing $\text{SUM}(\text{total_revenue_usd})$, $\text{SUM}(\text{profit_usd})$, $\text{SUM}(\text{order_count})$, and $\text{AVG}(\text{profit_margin_pct})$. The non-additive measure profit_margin_pct is explicitly aggregated with $\text{AVG}()$ — not SUM — in accordance with the measures glossary. This exploits the DFM design decision to store the raw percentage rather than forcing analysts to compute it from revenue and cost.

Revenue Over Time ($\text{dim_date} \times \text{FACT_SALES}$): A line chart of $\text{SUM}(\text{total_revenue_usd})$ and $\text{SUM}(\text{total_order})$ by full_date , demonstrating the **drill-down capability**: clicking a year shows months, clicking a month shows days. This is the core OLAP roll-up/drill-down operation supported by the date hierarchy in dim_date .

Quarterly Revenue ($\text{dim_date} \times \text{FACT_SALES}$): A bar chart aggregating $\text{SUM}(\text{total_revenue_usd})$ by **quarter**. Roll-up from the daily time series to quarterly periods illustrates how the **quarter** column in dim_date enables instant OLAP aggregation without computed fields.

3.3 Dashboard 2 — Product Performance

3.3.1 Purpose

Dashboard 2 answers: *Which products, categories, and brands generate revenue, and how are sub-categories distributed?* It fully exploits the three-level product hierarchy ($\text{category} \rightarrow \text{sub_category} \rightarrow \text{brand}$) encoded in dim_product .

3.3.2 Worksheets and Analytical Design

Revenue by Category ($\text{dim_product} \times \text{FACT_SALES}$): A horizontal bar chart of $\text{SUM}(\text{total_revenue_usd})$ by **category**. This is a roll-up to the top of the product hierarchy. The chart is designed for drill-down: clicking a category can filter the treemap and Top 10 chart to that category only, demonstrating the **dice** operation.

Top 10 Products ($\text{dim_product} \times \text{FACT_SALES}$): Ranked bar chart of product_name by $\text{SUM}(\text{total_profit_usd})$, limited to Top 10 via a Top N filter. A dual-axis shows $\text{SUM}(\text{quantity_sold})$, distinguishing high-revenue products from high-volume products — an insight requiring two additive measures simultaneously.

Sub-Category Treemap ($\text{dim_product} \times \text{FACT_SALES}$): Area-encoded treemap of $\text{SUM}(\text{total_revenue_usd})$ by **sub_category**. Colour encodes $\text{AVG}(\text{profit_margin_pct})$ (non-additive measure, correctly aggregated), revealing sub-categories where high revenue co-exists with low margins.

3.4 Dashboard 3 — Customer Segmentation

3.4.1 Purpose

Dashboard 3 answers: *How do customer segments, age groups, and loyalty levels differ in purchasing behaviour and revenue contribution?* It leverages `dim_customer` attributes that were specifically engineered during the ETL (age binning) and DQA (segment-median imputation of tenure).

3.4.2 Worksheets and Analytical Design

Revenue by Segment (`dim_customer` × `FACT_SALES`): A Pie chart of `SUM(total_revenue_usd)` grouped by `segment`.

Age Group Analysis (`dim_customer` × `FACT_SALES`): A stacked or grouped bar chart of revenue and order count by `age_group` (Young Adults, Adults, Mid-Career, Mature). This directly exploits the **Edit 1** DFM decision to bin `age` into four groups: without this transform, analysts would face 57 age values in the axis, making the chart unreadable.

Loyalty Score Distribution (`dim_customer` × `FACT_SALES`): A histogram or scatter chart of `loyalty_score` coloured by `segment` or `age_group`, showing how loyalty distributes across customer groups. This uses the continuous `loyalty_score` attribute from `dim_customer` — a dimension attribute retained as a numerical value rather than binned, allowing Tableau to compute `AVG(loyalty_score)` per segment.

3.5 Dashboard 4 — Geographic Distribution

3.5.1 Purpose

Dashboard 4 answers: *Where is revenue generated geographically, and which warehouses bear the highest fulfilment load?* It uses `dim_geography`, which merges the `COUNTRY` → `CITY` hierarchy with the warehouse location into a single flat dimension.

3.5.2 Worksheets and Analytical Design

World Map (`dim_geography` × `FACT_SALES`): A Tableau filled map where countries are colour-shaded by `SUM(total_revenue_usd)`. Tableau automatically resolves the `country` field in `dim_geography` to geographic coordinates. This is a **roll-up to country level**: the underlying city-level data in `dim_geography` allows drill-down to individual cities, demonstrating the OLAP hierarchy.

Country Ranking ($\text{dim_geography} \times \text{FACT_SALES}$): A horizontal ranked bar chart of countries by $\text{SUM}(\text{total_revenue_usd})$. A secondary bar shows $\text{SUM}(\text{total_order})$, highlighting countries where logistics costs represent a significant proportion of revenue.

Warehouse Heatmap ($\text{dim_geography} \times \text{FACT_SALES}$): A matrix heatmap of $\text{warehouse_location} \times \text{month_name}$ with colour intensity = $\text{SUM}(\text{order_count})$. This uses the order_count additive measure to show fulfilment load over time, identifying seasonality effects on each warehouse. The cross-dimension analysis ($\text{geography} \times \text{time}$) demonstrates **dice** across two dimensions.

3.6 Dashboard 5 — Marketing Effectiveness

3.6.1 Purpose

Dashboard 5 answers: *Which campaign sources and devices drive revenue, what are the main reasons customers return products, and how do these vary across segments?* It uses **dim_marketing** (campaign source, device type, traffic source) and **dim_order** (return reason — the MNAR column transformed to 'N/A' during ETL).

3.6.2 Worksheets and Analytical Design

Campaign Revenue ($\text{dim_marketing} \times \text{FACT_SALES}$): A bar or grouped chart of $\text{SUM}(\text{total_revenue_usd})$ and $\text{AVG}(\text{profit_margin_pct})$ by **campaign_source**. The non-additive measure is used correctly as $\text{AVG}()$ to compare margin quality across channels without distortion. This sheet enables **slicing by campaign source**: filtering to a single source updates all other dashboards via filter actions.

Return Reasons ($\text{dim_order} \times \text{FACT_SALES}$): A bar chart of $\text{SUM}(\text{order_count})$ by **return_reason**. The ETL decision to map $\text{NULL} \rightarrow \text{'N/A'}$ is essential here: without it, Tableau would either exclude NULLs from the chart or display them as blank labels. The 'N/A' value represents all non-returned orders and is the expected dominant bar, with actual return reasons occupying the remaining bars.

Device Revenue ($\text{dim_marketing} \times \text{FACT_SALES}$): A breakdown of $\text{SUM}(\text{total_revenue_usd})$ by **device_type** (Desktop, Mobile, Tablet) and **traffic_source**. This enables **slice-and-dice** across two marketing dimension attributes simultaneously, showing whether mobile traffic is as commercially effective as desktop.

Conclusions

This project delivered a complete, end-to-end data warehousing solution for an e-commerce sales dataset of 1,000,123 records across three phases:

The 62-column source dataset was normalised into a 12-table reconciled schema (3NF, PostgreSQL 18). A six-dimension ISO 25012 DQA achieved a 100% structural score. A five-method consensus outlier ensemble identified 8,883 rows (8.9%) for Winsorisation. Missingness analysis classified `return_reason` as MNAR and `tenure_days` as MAR, determining appropriate treatments for each. The cleaning pipeline applied 118,543 cell-level changes, raising accuracy from 98.01% to 99.67% without deleting a single row.

The Dimensional Fact Model (DFM) was constructed using the data-driven approach: fact choice, attribute tree construction, and five motivated editing steps (age binning, delivery binning, MNAR handling, hierarchy flattening, additivity classification). The resulting star schema (`ecommerce_dw`) contains 9 tables: `FACT_SALES` (991,930 rows) and 8 conformed dimension tables with 7 additive and 2 non-additive measures. The Python ETL pipeline passed post-load FK integrity and revenue sum reconciliation checks.

Five interactive Tableau dashboards covering Sales Overview, Product Performance, Customer Segmentation, Geographic Distribution, and Marketing Effectiveness were designed and implemented using 15 worksheets. OLAP operations — drill-down through the date hierarchy, slice by dimension, dice across multiple dimensions, and roll-up on product and geography hierarchies — are supported throughout. The non-additive measures (`profit_margin_pct`, `discount_pct`) are correctly aggregated with `AVG()` in all relevant sheets.